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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation represents the standard enthalpy change of atomisation of bromine?

- ☒ **A** $Br_2(g) \rightarrow 2Br(g)$
- ☒ **B** $Br_2(l) \rightarrow 2Br(g)$
- ☒ **C** $\frac{1}{2}Br_2(l) \rightarrow Br(g)$
- ☒ **D** $\frac{1}{2}Br_2(g) \rightarrow Br(g)$

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(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Chlorine reacts with ethane to form chloroethane and hydrogen chloride.

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$$Cl_2 + C_2H_6 \rightarrow C_2H_5Cl + HCl$$

Bond	Bond enthalpy / kJ mol^{-1}
C—Cl	346
C—H	413
Cl—Cl	243
H—Cl	432
C—C	347

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What is the enthalpy change for the reaction, in kJ mol^{-1}?

- ☒ **A** −1434
- ☒ **B** −122
- ☒ **C** +122
- ☒ **D** +1434

(Total for question = 1 mark)

Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Use the data shown.

$$CH_4(g) + 2F_2(g) \rightarrow CF_4(g) + 2H_2(g) \qquad \Delta H^{\ominus} = -858\,kJ\,mol^{-1}$$

$$C(s) + 2F_2(g) \rightarrow CF_4(g) \qquad \Delta H^{\ominus} = -933\,kJ\,mol^{-1}$$

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What is the standard enthalpy change of formation of methane (CH_4) in kJ mol^{-1}?

- ☒ **A** −1791
- ☒ **B** −75
- ☒ **C** +75
- ☒ **D** +1791

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(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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In an experiment, 50.0 cm^3 of 1.0 mol dm^{-3} HCl(aq) reacts with 50.0 cm^3 of 1.0 mol dm^{-3} NaOH(aq).

The energy released = 2500 J.

The specific heat capacity of the mixture is 4.18 J g^{-1} $°C^{-1}$

What temperature change occurs in the reaction?

- ☐ **A** an increase of 6.0°C
- ☐ **B** a decrease of 6.0°C
- ☐ **C** an increase of 12.0°C
- ☐ **D** a decrease of 12.0°C

(Total for question = 1 mark)

Q5.

Under certain conditions, graphite burns to form carbon monoxide.

$$2C(s) + O_2(g) \rightarrow 2CO(g) \qquad \Delta H = -221 \text{ kJ mol}^{-1}$$

Which of these is correct?

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(1)

- ☐ **A** $\Delta_c H$ (carbon) $= -221 \text{ kJ mol}^{-1}$
- ☐ **B** $\Delta_f H$ (carbon monoxide) $= -221 \text{ kJ mol}^{-1}$
- ☐ **C** $\Delta_c H$ (carbon) $= -110.5 \text{ kJ mol}^{-1}$
- ☐ **D** $\Delta_f H$ (carbon monoxide) $= -110.5 \text{ kJ mol}^{-1}$

(Total for question = 1 mark)

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Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The bond enthalpy for the Cl—Cl bond is +243.0 kJ mol^{-1}.

What is the enthalpy change of atomisation of chlorine in kJ mol^{-1}?

- ☐ **A** +243.0
- ☐ **B** –243.0
- ☐ **C** +121.5
- ☐ **D** –121.5

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(Total for question = 1 mark)

Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The standard enthalpy change of neutralisation for the reaction between sodium hydroxide solution and hydrochloric acid
is –56 kJ mol^{-1}.

Which row in the table is correct for this neutralisation?

	Reaction type	Temperature
☐ **A**	exothermic	increases
☐ **B**	exothermic	decreases
☐ **C**	endothermic	increases
☐ **D**	endothermic	decreases

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(Total for question = 1 mark)

Q8.

Magnesium reacts with sulfuric acid in an exothermic reaction.

(a) Write an equation for the reaction.

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Include state symbols in your answer.

(2)

(b) A student carried out an experiment to determine the enthalpy change of the reaction.

A sample of 0.50 g of magnesium powder was added to 25 cm^3 of 0.20 mol dm^{-3} sulfuric acid.
Calculate the number of moles of magnesium and of sulfuric acid that **reacted**.
Justify your answer.

(3)

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(c) (i) The results obtained are given in the table.

Plot the results on the grid.

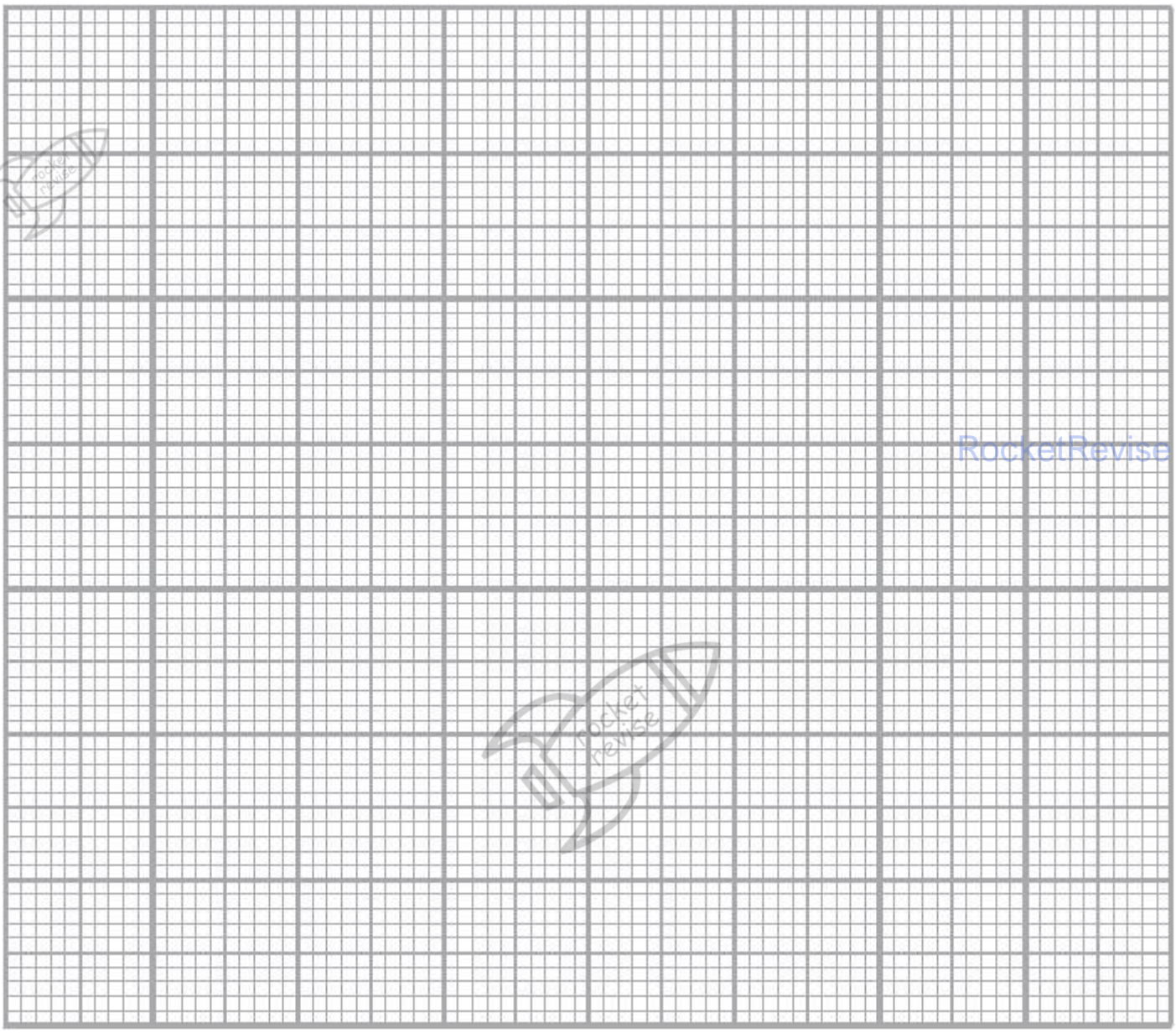
(2)

Time / minutes	Temperature / °C
0	22.2
1	22.0
2	22.0 ← Mg added
3	39.6
4	41.8
5	40.8
6	40.2
7	39.4
8	38.6

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(ii) Use your graph to determine the maximum change in temperature.
You **must** show your working on the graph.

(2)

ΔT = ...

(d) Calculate the standard molar enthalpy change for the reaction, using your answers to (b) and (c)(ii).
Include a sign and units in your answer.

[Specific heat capacity of solution = 4.18 J g^{-1} $°C^{-1}$]

(4)

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(Total for question = 13 marks)

Q9.

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This question is about enthalpy changes.

(a) An experiment was carried out to determine the enthalpy change of combustion for ethanol.

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$$

1.19 g of ethanol was burned in a spirit burner. The heat energy from this combustion raised the temperature of 100 g of water from 21.6 °C to 63.9 °C.

(i) Calculate the number of moles of ethanol in 1.19 g.
[Molar mass of ethanol = 46.0 g mol^{-1}]

(1)

(ii) Calculate the heat energy required to raise the temperature of 100 g of water from 21.6 °C to 63.9 °C.
[Specific heat capacity of water = 4.18 J g^{-1} $°C^{-1}$]

(2)

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(iii) Use your answers to (a)(i) and (ii) to calculate a value for the enthalpy change of combustion of ethanol.

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Give your answer to an appropriate number of significant figures and include a sign and units.

(3)

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(iv) The value of the enthalpy change of combustion from this experiment was very inaccurate. Give two reasons why this value was so inaccurate, apart from heat loss.

(2)

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(b) Mean bond enthalpies can be used to calculate a value for the enthalpy change of combustion of a compound.

(i) Give the meaning of the term 'mean bond enthalpy'.

(2)

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.....

.....

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(ii) Calculate a value for the enthalpy change of combustion of methanol, using the information in the table and the equation shown.

(3)

$$CH_3OH + 1\frac{1}{2}O_2 \rightarrow CO_2 + 2H_2O$$

	C—H	C—O	O—H	O=O	C=O
Mean bond enthalpy / kJ mol^{-1}	413	358	464	498	805

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(c) Enthalpy changes of combustion can be used to calculate the enthalpy change of formation of a compound.

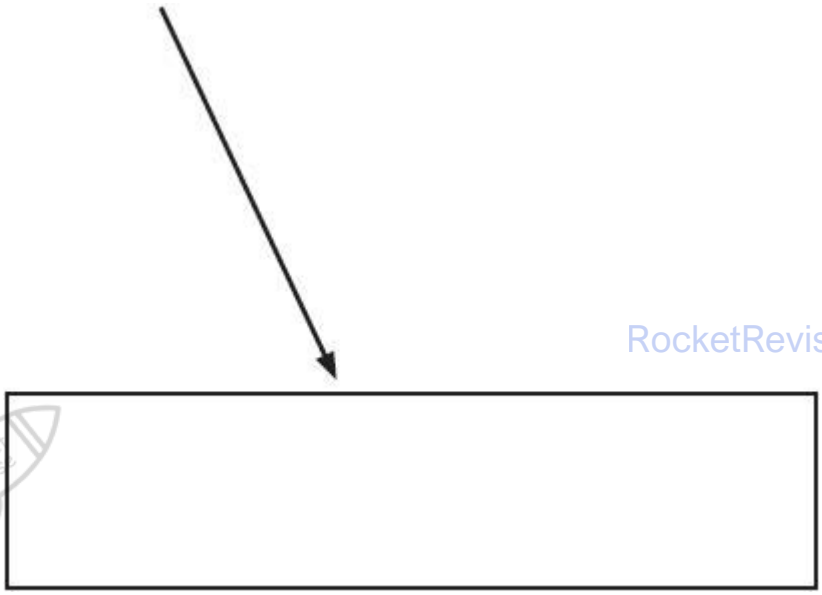
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Substance	Standard enthalpy change of combustion, $\Delta_c H^{\ominus}$ / kJ mol^{-1}
C(s,graphite)	−394
$H_2(g)$	−286
$CH_3CHO(g)$	−1167

Complete the Hess cycle and use it to calculate the standard enthalpy change of formation for ethanal, CH_3CHO.

(3)

$$2C(s, \text{graphite}) + 2H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow CH_3CHO(g)$$



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(Total for question = 16 marks)

Q10.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The equation for a reaction is

$$2C(s) + O_2(g) \rightarrow 2CO(g)$$

Which is the correct symbol for the enthalpy change for this reaction?

- ☐ **A** $\Delta_{at}H$
- ☐ **B** $\Delta_c H$
- ☐ **C** $\Delta_f H$
- ☐ **D** $\Delta_r H$

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(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The enthalpy changes for two reactions are shown.

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l) \qquad \Delta_r H = -57.2 \text{ kJ mol}^{-1}$$

$$CH_3COOH(aq) + OH^-(aq) \rightarrow CH_3COO^-(aq) + H_2O(l) \qquad \Delta_r H = -56.0 \text{ kJ mol}^{-1}$$

What is the enthalpy change for the dissociation of $CH_3COOH(aq)$ into $CH_3COO^-(aq)$ and $H^+(aq)$ ions, in kJ mol^{-1}?

- ☐ **A** +113.2
- ☐ **B** +1.2
- ☐ **C** −1.2
- ☐ **D** −113.2

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(Total for question = 1 mark)

Q12.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

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The table shows the amount of energy released per gram when some alkanes are burned in excess oxygen under standard conditions.

Alkane	Energy released / kJ g^{-1}
methane	55.6
ethane	52.0
propane	50.4
butane	49.6

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Which alkane has a standard enthalpy change of combustion of −2877 kJ mol^{-1}?

☒ **A** methane

☒ **B** ethane

☒ **C** propane

☒ **D** butane

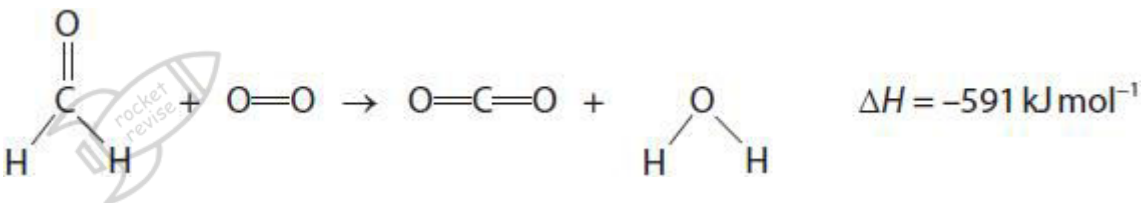
(Total for question = 1 mark)

Q13.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The equation for the complete combustion of methanal is shown.

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Some bond enthalpy data are shown.

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Bond	Bond enthalpy / kJ mol^{-1}
C—H	413
O═O	498
C═O in CO_2	805
O—H	464

What is the C═O bond enthalpy in methanal?

- ☒ **A** 623 kJ mol^{-1}
- ☒ **B** 678 kJ mol^{-1}
- ☒ **C** 805 kJ mol^{-1}
- ☒ **D** 1036 kJ mol^{-1}

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(Total for question = 1 mark)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Hydrogen reacts with oxygen to form steam.

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H2O(g) \qquad \Delta H = -242 \text{ kJ mol}^{-1}$$

Bond	Bond enthalpy / kJ mol^{-1}
H—H	436
O═O	498

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What is the bond enthalpy of the O—H bond, in kJ mol^{-1}?

- ☒ **A** 221.5
- ☒ **B** 463.5
- ☒ **C** 588
- ☒ **D** 927

(Total for question = 1 mark)

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Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the mass of carbon formed when 1.80 g of propan-1-ol, C_3H_7OH, undergoes incomplete combustion according to the equation shown?

$$C_3H_7OH(l) + 2O_2(g) \rightarrow 2C(s) + CO(g) + 4H_2O(l)$$

- ☒ **A** 0.18 g
- ☒ **B** 0.36 g
- ☒ **C** 0.72 g
- ☒ **D** 1.08 g

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(Total for question = 1 mark)

Q16.

A student carries out two experiments and uses the results to determine the enthalpy change when anhydrous copper(II) sulfate forms hydrated copper(II) sulfate, $CuSO_4 \cdot 5H_2O$.

$$CuSO_4(s) + 5H_2O(l) \rightarrow CuSO_4 \cdot 5H_2O(s)$$

(a) In the first experiment, the student determines the enthalpy change when hydrated copper(II) sulfate dissolves in water.

$$CuSO_4 \cdot 5H_2O(s) + aq \rightarrow CuSO_4(aq)$$

10.68 g of hydrated copper(II) sulfate is added to distilled water in a polystyrene cup to give 55.0 g of solution.
The mixture is stirred and the temperature change determined.

Results

Initial temperature = 21.0 °C
Minimum temperature = 18.5 °C

Calculate the enthalpy change for this reaction.
Give your answer to an appropriate number of significant figures.
Include a sign and units.

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Data: Specific heat capacity of the solution = 3.70 J g^{-1} $°C^{-1}$

Molar mass $CuSO_4 \cdot 5H_2O$ = 249.6 g mol^{-1}

(4)

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(b) In the second experiment, the student determined the enthalpy change of the reaction when anhydrous copper(II) sulfate dissolves in water.

$$CuSO_4(s) + aq \rightarrow CuSO_4(aq) \qquad \Delta H = -67.4 \text{ kJ mol}^{-1}$$

(i) Complete the Hess cycle.

(2)



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(ii) Calculate the enthalpy change when anhydrous copper(II) sulfate forms hydrated copper(II) sulfate. Include a sign and units.

(2)

(Total for question = 8 marks)

Q17.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation represents the standard enthalpy change of formation, $\Delta_f H^{\ominus}$, for aluminium oxide?

- ☐ **A** $2Al(s) + 1\frac{1}{2}O_2(g) \rightarrow Al_2O_3(s)$
- ☐ **B** $2Al(s) + 3O(g) \rightarrow Al_2O_3(s)$
- ☐ **C** $4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$
- ☐ **D** $4Al(s) + 6O(g) \rightarrow 2Al_2O_3(s)$

(Total for question = 1 mark)

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Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some enthalpy changes of combustion are shown.

Substance	Enthalpy change of combustion / kJ mol^{-1}
$CH_3CHO(l)$	−1167
C(s)	−394
$H_2(g)$	−286

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What is the enthalpy change of formation of ethanal, CH_3CHO, in kJ mol^{-1}?

- ☒ **A** +765
- ☒ **B** −765
- ☒ **C** +193
- ☒ **D** −193

(Total for question = 1 mark)

Q19.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

5.20 g of sodium hydrogencarbonate is added to an excess of acid.

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The temperature increases and the energy change is calculated to be 1030 J.

What is the enthalpy change per mole of sodium hydrogencarbonate?

[M_r $NaHCO_3$ = 84.0]

- ☒ **A** –12.3 kJ mol^{-1}
- ☒ **B** –16.6 kJ mol^{-1}
- ☒ **C** –63.8 kJ mol^{-1}
- ☒ **D** –16 600 kJ mol^{-1}

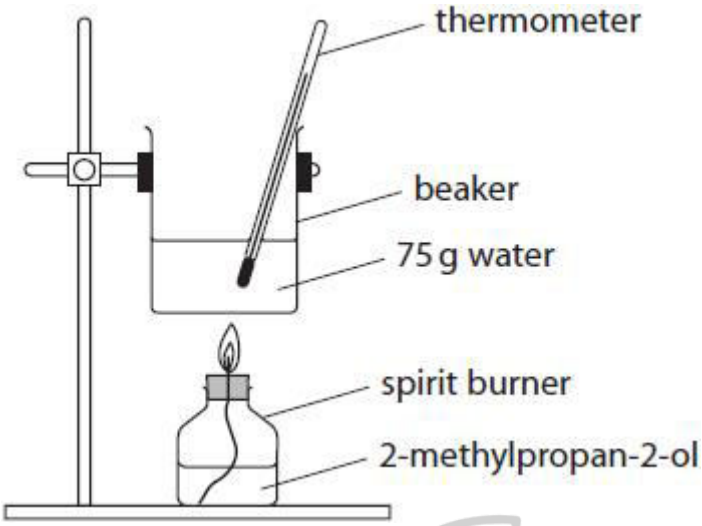
(Total for question = 1 mark)

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Q20.

Enthalpy changes of combustion can be determined using calorimetry or calculated using Hess cycles.
Apparatus for a calorimetry experiment is shown.



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A sample of 2-methylpropan-2-ol was burned in a spirit burner and used to heat 75 g of water. The results are shown.

	At the start	At the end	Change
Mass of spirit burner / g	267.35	266.78	
Temperature of water / °C	19.5	65.3	

(a) (i) Complete the table.

(1)

(ii) Calculate the enthalpy change of combustion, $\Delta_c H$, of 2-methylpropan-2-ol.
Give a sign and units in your answer.
[Specific heat capacity of water = 4.18 J g^{-1} $°C^{-1}$]

(4)

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(b) The standard enthalpy change of combustion, $\Delta_c H^{\ominus}$, can be calculated using standard enthalpy changes of formation.

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Compound	$\Delta_f H^{\ominus}$ / kJ mol^{-1}
2-methylpropan-2-ol	−359
carbon dioxide	−394
water	−286

(i) State why no $\Delta_f H^{\ominus}$ value has been given for oxygen.

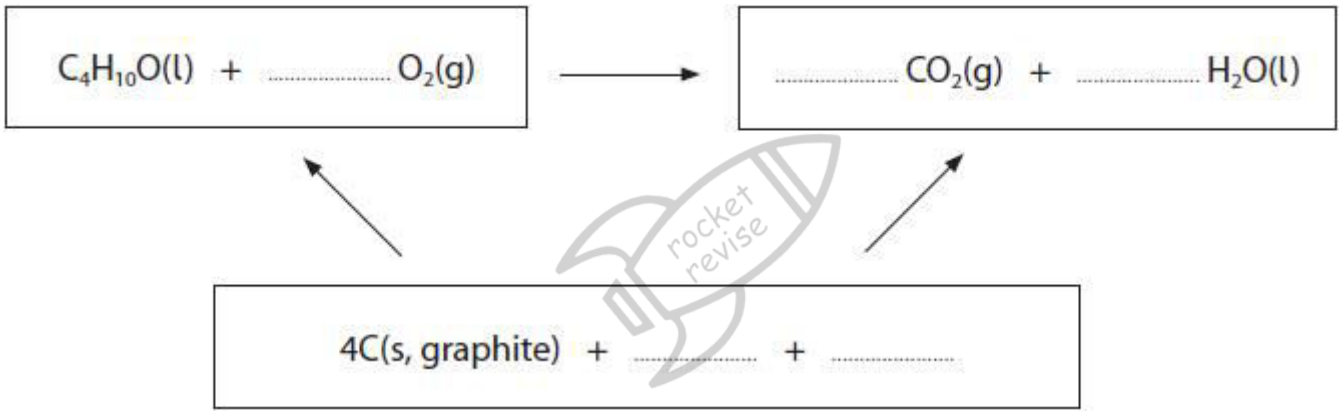
(1)

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(ii) Complete the Hess cycle.



(2)

(iii) Calculate the standard enthalpy change of combustion of 2-methylpropan-2-ol using the data in the table and the completed Hess cycle.

(2)

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(c) The value for $\Delta_c H$ obtained in part (a)(ii) is much less exothermic than $\Delta_c H^{\ominus}$

Suggest **two** reasons for this other than non-standard conditions.

(2)

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(Total for question = 12 marks)

Q21.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which are correct for the reaction shown?

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$CH_3COOH + KOH \rightarrow CH_3COO^-K^+ + H_2O \quad \Delta H = -55.8 \text{ kJ mol}^{-1}$

	Type of reaction	Type of enthalpy change
☐ **A**	endothermic	formation
☐ **B**	endothermic	neutralisation
☐ **C**	exothermic	formation
☐ **D**	exothermic	neutralisation

(Total for question = 1 mark)

Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation shows the standard enthalpy change of formation of sodium chloride?

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- ☐ **A** $Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$
- ☐ **B** $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$
- ☐ **C** $Na(g) + Cl(g) \rightarrow NaCl(s)$
- ☐ **D** $Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$

(Total for question = 1 mark)

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Q23.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation does **not** represent a standard enthalpy change of atomisation?

- ☐ **A** $Mg(s) \rightarrow Mg(g)$
- ☐ **B** $Cl_2(g) \rightarrow 2Cl(g)$
- ☐ **C** $\frac{1}{2}O_2(g) \rightarrow O(g)$
- ☐ **D** $Hg(l) \rightarrow Hg(g)$

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(Total for question = 1 mark)

Q24.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The mean C–F bond enthalpy is +485 kJ mol^{-1}.

Which process has an enthalpy change of +1940 kJ mol^{-1}?

- ☐ **A** $C(g) + 4F(g) \rightarrow CF_4(g)$
- ☐ **B** $C(s) + 2F_2(g) \rightarrow CF_4(g)$
- ☐ **C** $CF_4(g) \rightarrow C(g) + 4F(g)$
- ☐ **D** $CF_4(g) \rightarrow C(s) + 2F_2(g)$

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(Total for question = 1 mark)

Q25.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Excess zinc powder is added to 50.00 g of 0.500 mol dm^{-3} copper(II) sulfate solution in a polystyrene cup. The mixture is stirred and the maximum temperature change determined.

The enthalpy change for the reaction is calculated to be −185 kJ mol^{-1}.

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The data book value for this reaction is −217 kJ mol^{-1}.

(a) The percentage error in this experiment is

(1)

- ☒ **A** ±7.37 %
- ☒ **B** ±8.65 %
- ☒ **C** 14.7 %
- ☒ **D** 17.3 %

(b) In the calculation, the specific heat capacity of the liquid is taken to be 4.18 J g^{-1} °C^{-1} rather than the true value of 3.85 J g^{-1} °C^{-1}.

The use of 4.18 J g^{-1} °C^{-1} in the calculation

(1)

- ☒ **A** is partly responsible for the error in the final value obtained
- ☒ **B** lowers the error in the final value obtained
- ☒ **C** has a negligible effect on the final value obtained
- ☒ **D** has a negligible effect compared with the measurement uncertainties

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(Total for question = 2 marks)

Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The enthalpy change of reaction, $\Delta_r H$, for the equation shown can be calculated using bond enthalpy data.

$$\tfrac{1}{2}H_2(g) + \tfrac{1}{2}Cl_2(g) \rightarrow HCl(g)$$

Bond	Bond enthalpy / kJ mol^{-1}
H—H	436
Cl—Cl	242
H—Cl	431

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The expression that should be used in the calculation is

- ☒ **A** (0.5 × 436 + 0.5 × 242) – 431
- ☒ **B** (2 × 431) – (436 + 242)
- ☒ **C** 431 – (0.5 × 436 + 0.5 × 242)
- ☒ **D** (436 + 242) – (2 × 431)

(Total for question = 1 mark)

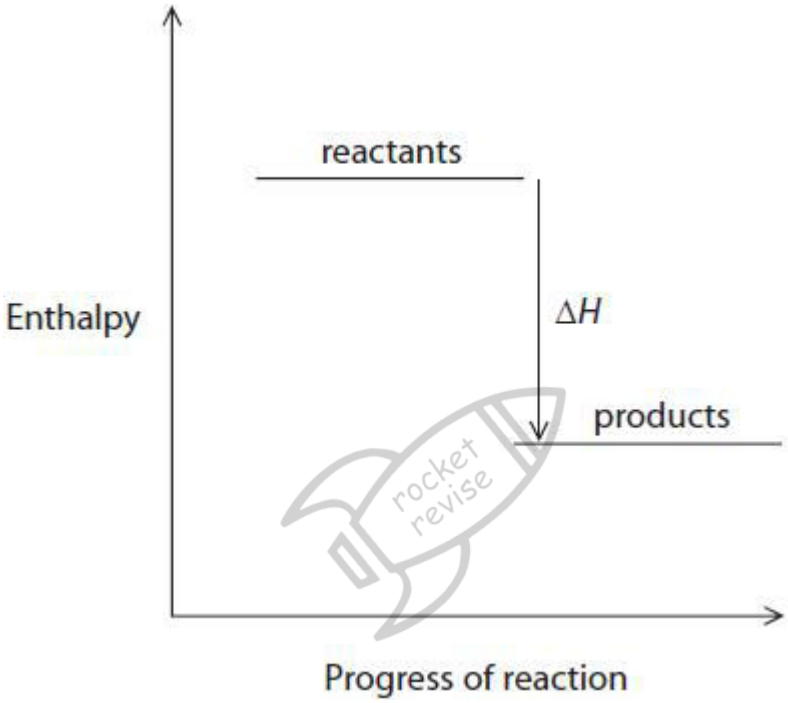
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Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The enthalpy level diagram for a reaction is shown.



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Which is represented by this diagram?

- ☒ **A** $Na(s) \rightarrow Na(g)$
- ☒ **B** $Na(g) \rightarrow Na^+(g) + e^-$
- ☒ **C** $2Cl(g) \rightarrow Cl_2(g)$
- ☒ **D** $NH_4NO_3(s) + aq \rightarrow NH_4^+(aq) + NO_3^-(aq)$

(Total for question = 1 mark)

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Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The enthalpy changes for two reactions are shown.

$$2Fe(s) + 1\tfrac{1}{2}O_2(g) \rightarrow Fe_2O_3(s) \qquad \Delta_r H^{\ominus} = -824\,kJ\,mol^{-1}$$

$$C(s) + \tfrac{1}{2}O_2(g) \rightarrow CO(g) \qquad \Delta_r H^{\ominus} = -111\,kJ\,mol^{-1}$$

What is the enthalpy change for the reaction between iron(III) oxide and carbon?

$$Fe_2O_3(s) + 3C(s) \rightarrow 2Fe(s) + 3CO(g)$$

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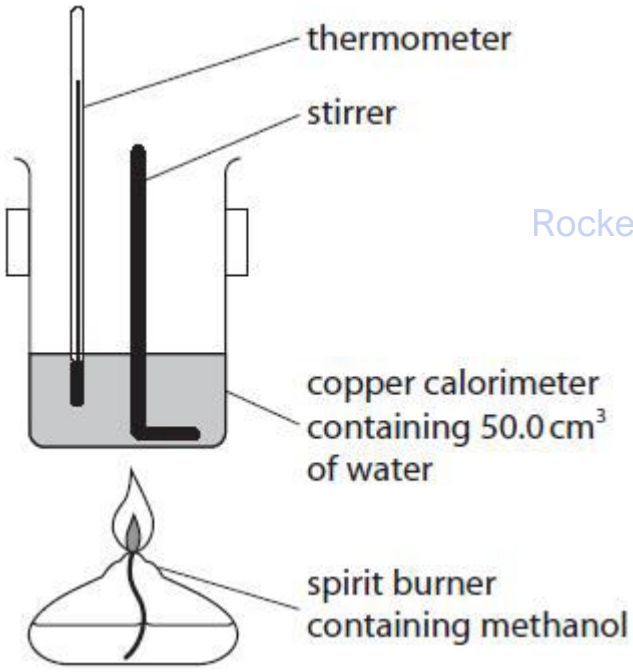
- ☐ **A** +491 kJ mol^{-1}
- ☐ **B** +713 kJ mol^{-1}
- ☐ **C** −491 kJ mol^{-1}
- ☐ **D** −713 kJ mol^{-1}

(Total for question = 1 mark)

Q29.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student measures the enthalpy change of combustion, $\Delta_c H$, of methanol, CH_3OH , using the apparatus shown.



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After burning 0.20 g of methanol, the temperature of the water increases by 16.0 °C .

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(a) The measurement uncertainty in the thermometer used in the experiment is 0.5 °C for each reading.

What is the percentage uncertainty in the temperature change of 16.0 °C ?

(1)

- ☐ **A** 1.6%
- ☐ **B** 3.1%
- ☐ **C** 6.3%
- ☐ **D** 12.5%

(b) The student repeats the experiment but burns 0.30 g of methanol and uses 75.0 cm^3 of water in the copper calorimeter.

What is the expected temperature change in this repeat experiment?

(1)

- ☐ **A** 7.1 °C
- ☐ **B** 10.7 °C
- ☐ **C** 16.0 °C
- ☐ **D** 36.0 °C

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(c) The student's calculated enthalpy change of combustion of methanol is **more** exothermic than a data book value.

What is a possible reason for this?

(1)

- ☐ **A** heat loss to the surroundings
- ☐ **B** incomplete combustion of methanol
- ☐ **C** evaporation of methanol from the wick of the spirit burner
- ☐ **D** use of the molar mass of ethanol, C_2H_5OH , in the calculation

(Total for question = 3 marks)

Q30.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which expression gives the standard enthalpy change, in kJ mol^{-1}, for the reaction shown?

$$BaCO_3(s) \rightarrow BaO(s) + CO_2(g)$$

$\Delta_f H^{\ominus}$ values: $BaCO_3(s) = -1216\,kJ\,mol^{-1}$

$BaO(s) = -554\,kJ\,mol^{-1}$

$CO_2(g) = -394\,kJ\,mol^{-1}$

- ☐ **A** –554 – 394 + 1216
- ☐ **B** –554 – 394 – 1216
- ☐ **C** 554 + 394 + 1216
- ☐ **D** 554 + 394 – 1216

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(Total for question = 1 mark)

Q31.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The standard enthalpy changes of combustion for a series of alkanes are shown.

Alkane formula	$\Delta_c H^{\ominus}$ / kJ mol^{-1}
$C_2H_6(g)$	−1560
$C_3H_8(g)$	−2219
$C_4H_{10}(l)$	−2877
$C_5H_{12}(l)$	−3509

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Another alkane has an enthalpy change of combustion of −6125 kJ mol^{-1}.

Which is the most likely formula for this alkane?

- ☐ **A** C_6H_{14}
- ☐ **B** C_7H_{16}
- ☐ **C** C_8H_{18}
- ☐ **D** C_9H_{20}

(Total for question = 1 mark)

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Q32.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation represents the standard enthalpy change of atomisation, $\Delta_{at}H^{\ominus}$, of bromine?

- ☐ **A** $\frac{1}{2}Br_2(l) \rightarrow Br(g)$
- ☐ **B** $\frac{1}{2}Br_2(g) \rightarrow Br(g)$
- ☐ **C** $Br_2(l) \rightarrow 2Br(g)$
- ☐ **D** $Br_2(g) \rightarrow 2Br(g)$

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(Total for question = 1 mark)

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Q33.

Sodium hydrogencarbonate can be decomposed to sodium carbonate by heating to about 300 °C.

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The equation for the reaction is:

$$2NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(l)$$

(a) Give a reason why it is **not** possible to measure the enthalpy change for this reaction directly.

(1)

.....

.....

.....

(b) (i) State what is meant by the standard enthalpy change of formation.

(2)

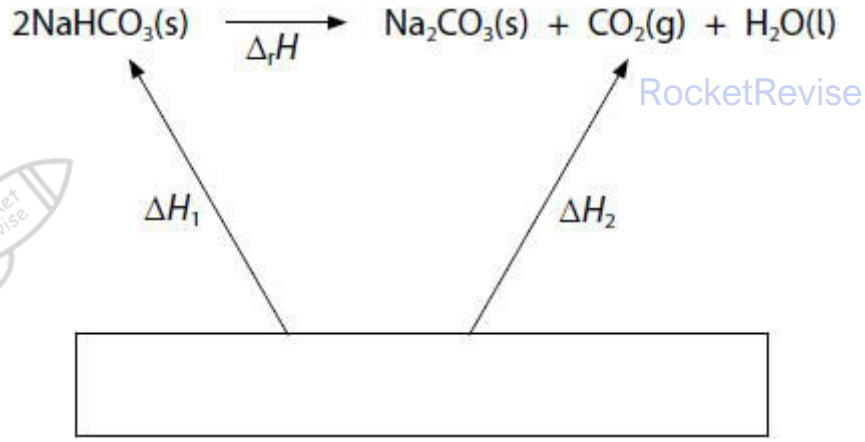
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(ii) Complete the Hess cycle that you would use to determine the enthalpy change for this reaction from the standard enthalpy changes of formation.

(2)



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(iii) Calculate the standard enthalpy change for the thermal decomposition of sodium hydrogencarbonate, using the information in the table and your completed cycle. Include a sign and units in your answer.

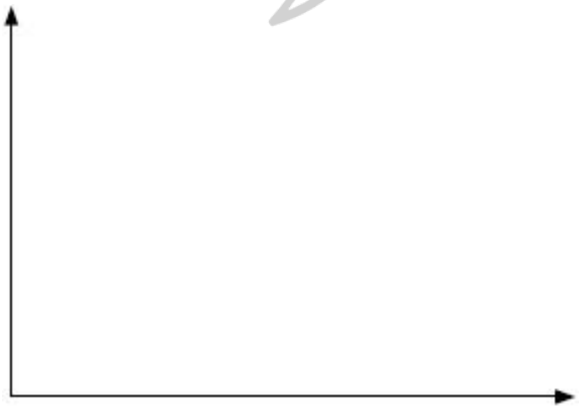
(4)

Compound	Standard enthalpy change of formation, $\Delta_f H^{\ominus}$ / kJ mol^{-1}
$NaHCO_3(s)$	−950.8
$Na_2CO_3(s)$	−1130.7
$CO_2(g)$	−393.5
$H_2O(l)$	−285.8

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(iv) Use your answer to (b)(iii) to draw an enthalpy level diagram for this reaction, labelling the axes provided.

(2)



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(Total for question = 11 marks)

Q34.

This question is about enthalpy changes.

A student wanted to use Hess's Law to determine the enthalpy change, $\Delta_r H_1$, in kJ mol^{-1}, for Reaction **1**.

Reaction **1** $\quad Mg(s) + \frac{1}{2}O_2(g) \rightarrow MgO(s) \quad \Delta_r H_1$

(a) Explain why $\Delta_r H_1$ can also be described as an enthalpy change of formation.

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(2)

.....

.....

.....

(b) The student carried out an experiment to determine the enthalpy change of reaction, $\Delta_r H_2$, in kJ mol^{-1}, for Reaction **2**.

Reaction **2** $MgO(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l)$ $\Delta_r H_2$

0.189 g of solid magnesium oxide was added to excess hydrochloric acid in a polystyrene cup and stirred until the reaction was complete.
The total mass of the solution formed was 25.0 g.
During the experiment, the temperature increased from 21.5°C to 28.0°C.
Calculate $\Delta_r H_2$, in kJ mol^{-1}, including a sign.
Give your answer to an appropriate number of significant figures.
[Specific heat capacity of the solution = 4.18 J g^{-1}° C^{-1}.]

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(4)

(c) Calculate the enthalpy change of reaction, $\Delta_r H_1$, for Reaction **1**.

Use your answer to (b), the data provided and the enthalpy cycle shown.
[If you did not calculate an answer to (b), use a value of −100 kJ mol^{-1}. This is **not** the correct answer.]

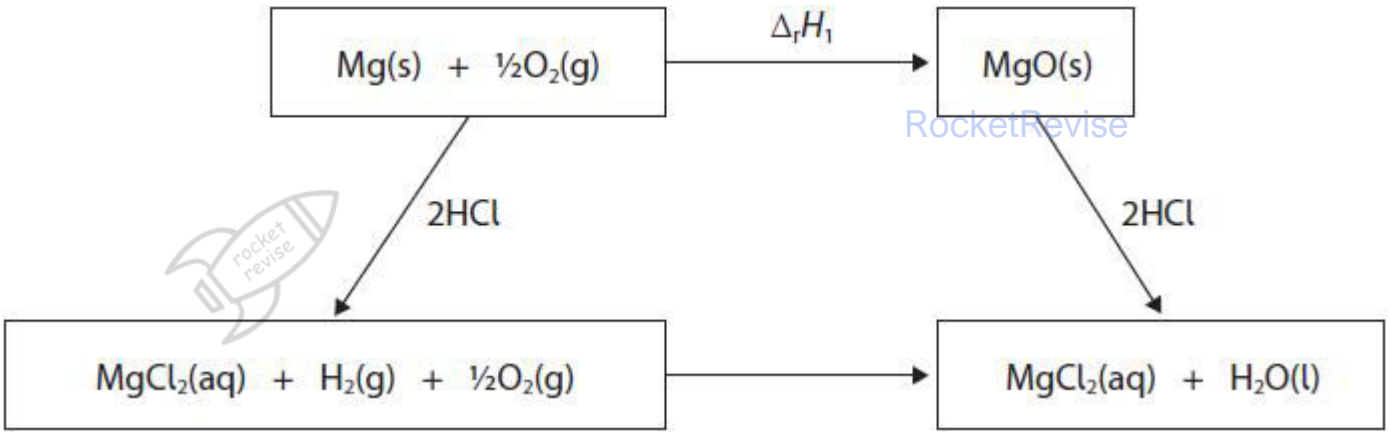
(2)

Data

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$ $\Delta_r H = -462\ kJ\ mol^{-1}$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ $\Delta_f H = -286\ kJ\ mol^{-1}$

Enthalpy cycle



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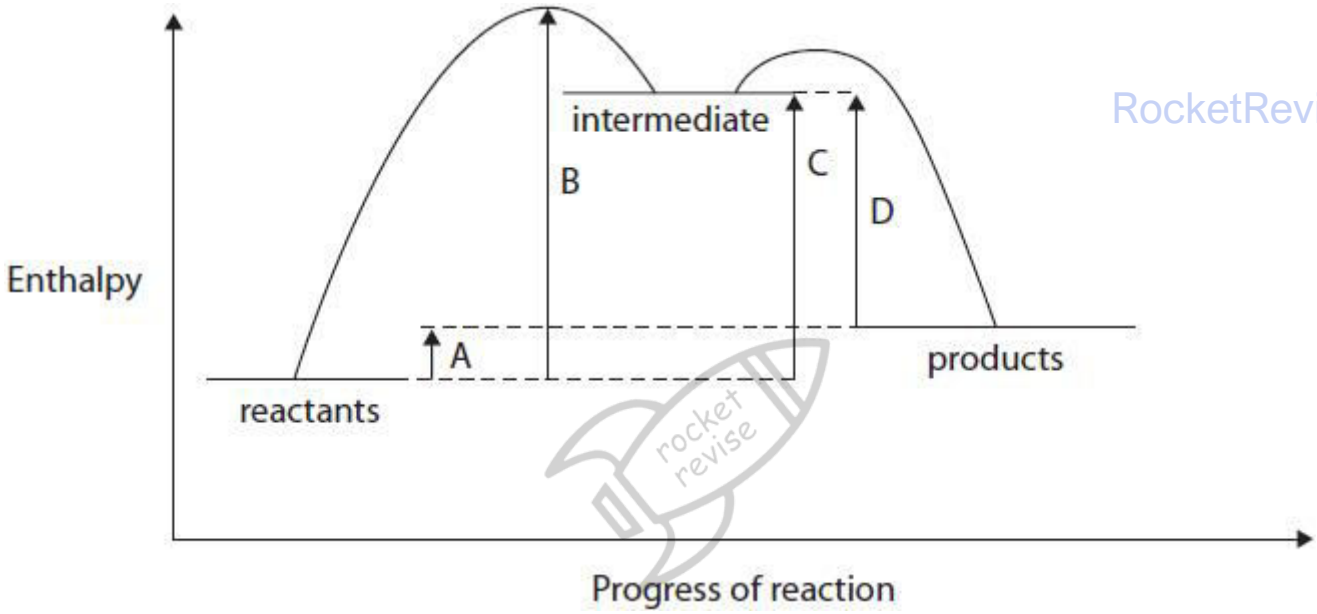
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(Total for question = 8 marks)

Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An enthalpy level diagram for a reaction with an intermediate is shown.



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Which arrow shows the enthalpy change for this reaction?

- ☐ **A** arrow A
- ☐ **B** arrow B
- ☐ **C** arrow C
- ☐ **D** arrow D

(Total for question = 1 mark)

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Q36.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation represents the reaction for which the enthalpy change, $\Delta_r H$, is the mean bond enthalpy of the C—Cl bond in CCl_4?

- ☐ **A** $CCl_4(g) \rightarrow C(g) + 4Cl(g)$
- ☐ **B** $\frac{1}{4}CCl_4(g) \rightarrow \frac{1}{4}C(g) + Cl(g)$
- ☐ **C** $C(g) + 4Cl(g) \rightarrow CCl_4(g)$
- ☐ **D** $\frac{1}{4}C(g) + Cl(g) \rightarrow \frac{1}{4}CCl_4(g)$

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(Total for question = 1 mark)

Q37.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which equation shows the reaction that occurs when the standard enthalpy change of formation of copper(II) carbonate is measured?

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- ☒ **A** $Cu(s) + C(s) + 3O(g) \rightarrow CuCO_3(s)$
- ☒ **B** $2Cu(s) + 2C(s) + 3O_2(g) \rightarrow 2CuCO_3(s)$
- ☒ **C** $Cu(g) + C(g) + 1½O_2(g) \rightarrow CuCO_3(s)$
- ☒ **D** $Cu(s) + C(s) + 1½O_2(g) \rightarrow CuCO_3(s)$

(Total for question = 1 mark)

Q38.

In a titration between a weak acid and a weak base indicators do not give a distinct colour change, so other techniques must be used.
One technique is thermometric titration.

Procedure

Step **1** The temperature of 30.0 cm^3 of dilute ethanoic acid in a polystyrene cup is recorded.

Step **2** Ammonia solution of concentration 1.30 mol dm^{-3} is added to the acid in the polystyrene cup in separate 5.00 cm^3 portions.

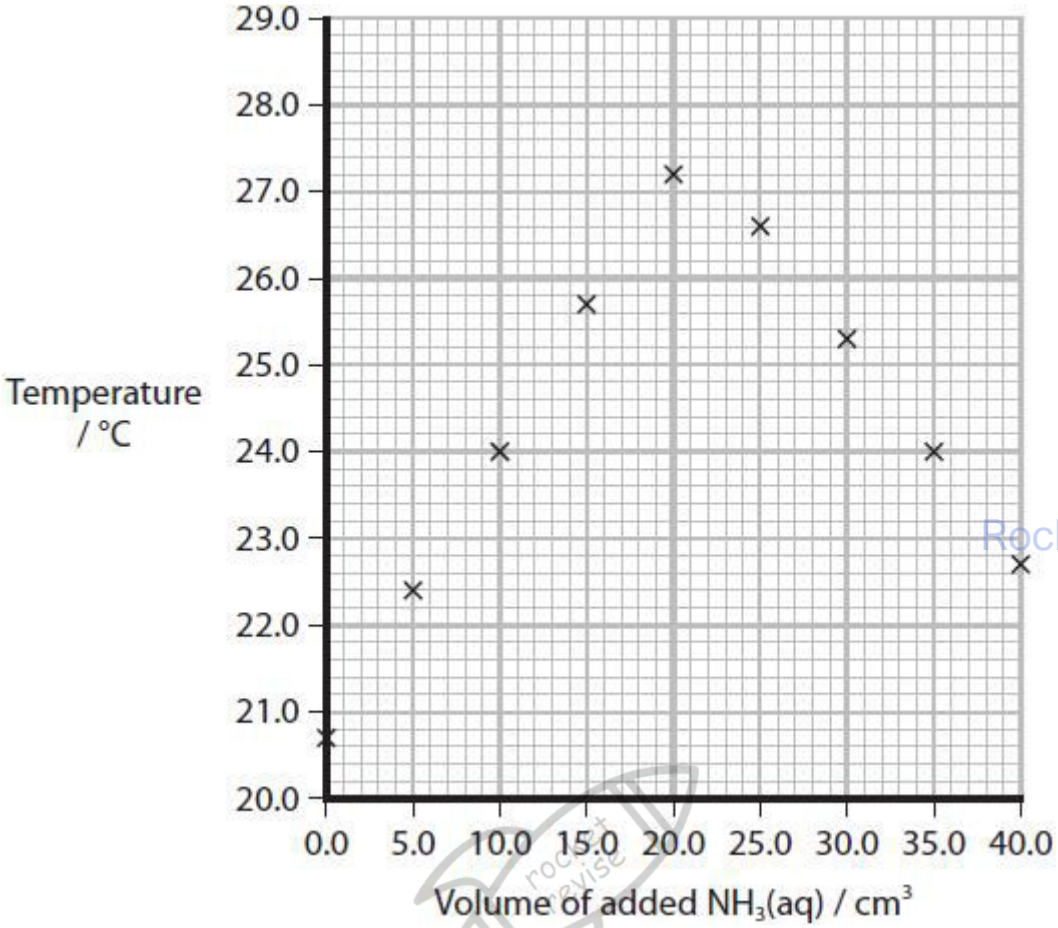
After each addition the mixture is stirred and the temperature measured.

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A student carried out the experiment and plotted the graph shown.

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(a) Find the end-point volume and the maximum temperature by adding appropriate lines of best fit to the graph.

(3)

End-point volume = ...

Maximum temperature = ...

(b) Calculate the number of moles of ammonia added at the end-point.

(1)

(c) The reaction that occurs is shown.

$$CH_3COOH(aq) + NH_3(aq) \rightarrow CH_3COONH_4(aq)$$

Calculate the enthalpy change of neutralisation for this reaction, in kJ mol^{-1}. Give your answer to an appropriate number of significant figures and include a sign with your answer.
[Assume that the density of all solutions is 1.00 g cm^{-3} and that the specific heat capacity of all solutions is 4.18 J g^{-1} $°C^{-1}$]

(5)

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(Total for question = 9 marks)

Q39.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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(a) Methane burns to produce carbon dioxide and water vapour.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g) \quad \Delta H = -810 \text{ kJ mol}^{-1}$$

Some bond enthalpy data are shown.

Bond	Bond enthalpy / kJ mol^{-1}
C—H	413
O=O	498
O—H	464

What is the mean bond enthalpy for the C≡O bond, in kJ mol^{-1}?

(1)

- ☒ **A** 552
- ☒ **B** 801
- ☒ **C** 1016
- ☒ **D** 1265

(b) A data book value for the standard enthalpy change of combustion of methane is −890 kJ mol^{-1}.

What is the experimental value for the enthalpy change of vaporisation of water in kJ mol^{-1}?

$$H_2O(l) \rightarrow H_2O(g)$$

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(1)

- ☒ **A** −40
- ☒ **B** +40
- ☒ **C** −80
- ☒ **D** +80

(Total for question = 2 marks)

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